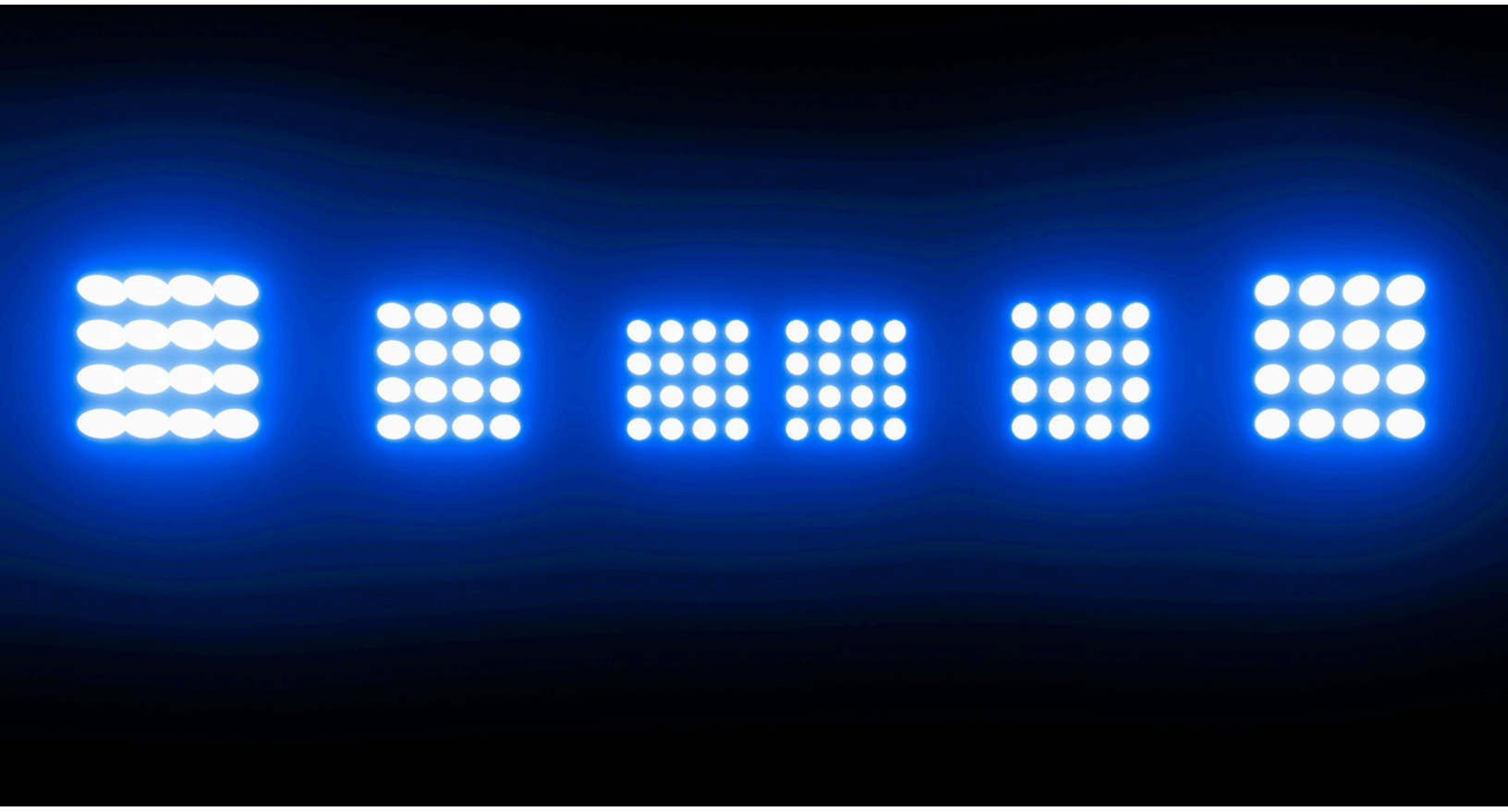
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Capturing the next big arenas of competition in ten charts

Arenas are industries that transform the business landscape. As 18 future arenas emerge, companies should understand their own exposure and positioning now.

by Chris Bradley, Kevin Russell, Kweilin Ellingrud, and Suhayl Chettih



November 2025

In every era, promising pockets of the industrial landscape emerge to transform the economy. These burgeoning "arenas of competition"—"arenas," for short—are industries with the fastest growth and the most competitive dynamism.

Our 2024 research, which identified arenas' core characteristics of high growth and dynamism, found 12 such industries—a group that included cloud services, e-commerce, biopharmaceuticals, and electric vehicles (EVs). This dynamic dozen more than doubled their revenue share between 2005 and 2023 and grew their market capitalization by 14 percent per year, compared with 5 percent for the 45 other (non-arena) industries analyzed. The 12 arenas saw more new entrants, much higher R&D investment, and greater economic profits, too. At the root of their growing success, we found an "arena-creation potion" with three main ingredients: a technology or business model breakthrough, an escalatory race with ever-larger investments, and an addressable market that enables global scale and is large, growing, or both. In short, arenas are characterized by a particularly intense race to win, with outsize rewards but also a high risk of displacement.

Why does this matter? We see that the arena-creation potion is already at work transforming 18 additional industries that may evolve into the next big arenas of competition over the next 15 years—from semiconductors, AI software and services, and cybersecurity, to more physical realms like space, robotics, and modular construction. If the past is any guide, these 18 arenas will be tomorrow's centers of competition, innovation, and value creation. For companies, tracking current and emerging arenas may show where next to compete, how to transform operations, and where downstream demand might grow quickly. Tracking arenas also makes sense for investors aiming to maximize their returns, job seekers searching for productive careers, and policymakers looking to play a role in how and where these industries develop.

This article spotlights the flagship McKinsey Global Institute report, first published in October 2024, in ten condensed segments, with a conclusion that reflects new company-level applications based on insights gleaned from a year's experience helping companies prepare to compete in these next big arenas.

Arenas are characterized by a particularly intense race to win, with outsize rewards but also a high risk of displacement.

1

The power of arenas

The industrial landscape has shifted dramatically over the past 20 years. Just look at the top ten most valuable companies in 2005 and 2025. Only one company appears on both lists. And the rest of the 2025 leaders are worth about ten times more than the 2005 leaders they replaced (Exhibit 1).

What has caused this radical reshuffling? And why are today's winners winning on a whole new scale? We investigated the forces in play and how they might explain why some industries clearly create more value and have a greater impact than others.

Exhibit 1

The past 20 years have seen a radical reshuffling in the ranking of the top ten companies.

Company ranking by market cap, $ billion

Arenas of today ○ New entries

Ranking, 2005			**Ranking,** 2025[1]		
1	General Electric	370	1	Nvidia	5,027
2	ExxonMobil	350	2	Apple	3,976
3	Microsoft	278	3	Microsoft	3,843
4	Citi	246	4	Alphabet	3,426
5	BP	229	5	Amazon	2,715
6	Shell	211	6	Broadcom	1,712
7	Walmart	195	7	Aramco	1,650
8	Bank of America	185	8	Meta	1,607
9	Johnson & Johnson	179	9	Tesla	1,558
10	HSBC	177	10	TSMC	1,267
Total market cap, $ billion		*2,420*			*26,780*

[1]Data as of November 3, 2025.
Source: McKinsey Value Intelligence; McKinsey Global Institute analysis

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2

Twelve arenas have emerged over the past two decades

The 12 big arenas of today showed outsize growth and market dynamism between 2005 and 2023. Represented with light blue circles, the 12 industries are cloud services, EVs, payments, e-commerce, consumer electronics, biopharmaceuticals, video and audio entertainment, industrial electronics, semiconductors, consumer internet, software, and information-enabled business services (Exhibit 2). Each one dramatically outpaced overall economic growth over the past two decades, as captured in the change in industry's share of total market cap metric. They also saw competitors' positions change dramatically within their industries, as captured in the "shuffle rate," measured as the cumulative market share shifts among companies. To select the 12 arenas from the 57 industries analyzed, we ranked all of them according to their combined growth and dynamism scores across both revenue and market cap, filtering out any that fell below our 2023 market cap threshold of $800 billion.

Exceptional growth put the 12 arenas to the right of our chart. Collectively, the 12 arenas nearly tripled their share of total market capitalization from 2005 to 2023. Tech-driven categories—including cloud services, e-commerce, and consumer internet—saw the largest industry shifts. For example, e-commerce had one of the largest growth rates for revenues, jumping from $15 billion in 2005 to more than $1 trillion in 2023.

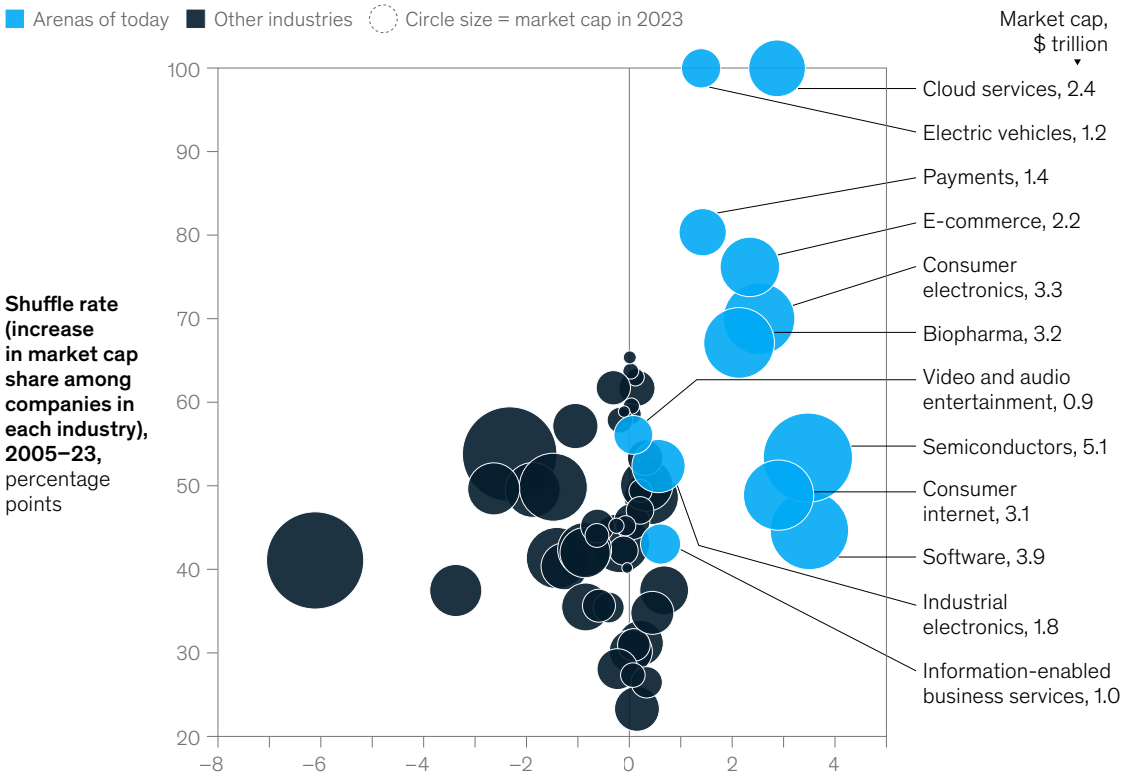
Elevated dynamism makes arenas stand out, too. The average market-cap shuffle rate was 66 percent for arenas, compared with 45 percent for other industries. Meanwhile, the average revenue shuffle rate for arenas was 53 percent, compared with 34 percent for other industries.

Each one of the 12 big arenas of today has dramatically outpaced overall economic growth over the past two decades.

Exhibit 2

The 12 arenas of today exhibited outsize shuffle rates and significant growth in share by market cap.

Market cap shuffle rates and growth share for 57 industries



Note: Exhibit and analysis updated to 2023. Source: McKinsey Value
Source: McKinsey Value Intelligence; McKinsey Global Institute analysis

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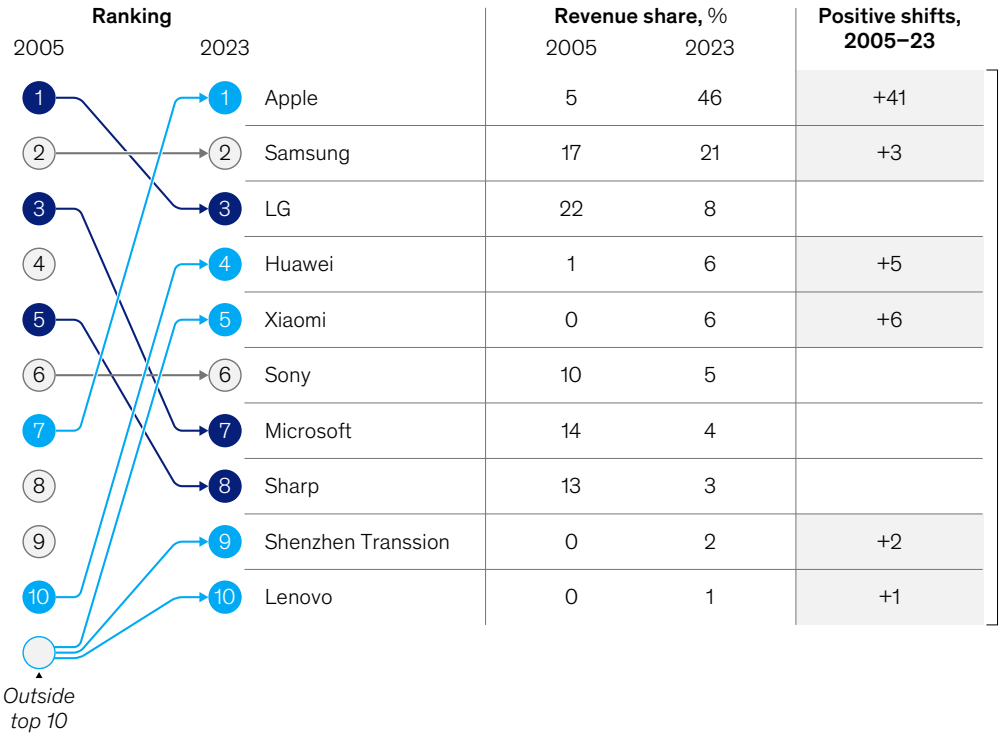
In arenas, companies engage in highly dynamic competition

To see how shuffle rates capture an arena's dynamism, consider consumer electronics. Here, Apple grew its share of industry revenues from 5 percent in 2005 to 46 percent in 2023, while Xiaomi, which entered the market in 2010, scaled up to capture 6 percent of the market by 2023, reflecting significant reshuffling within the industry. Summing all positive market share gains across consumer electronics companies yielded a shuffle rate of 57 percentage points for that industry (Exhibit 3). In contrast, aerospace and defense had a much lower shuffle rate of 15 percentage points, indicating less dynamism among its players in the same period. In nascent industries like EVs and cloud services, revenues and market caps first emerged, and shuffle rates were set at 100 percent.

Exhibit 3

The top ten companies in the consumer electronics arena had a 57-point shuffle rate.

Company ranking by market share of revenue



Note: Exhibit and analysis updated to 2023.
[1]Calculated by summing all positive market share percentage-point changes between 2005 and 2023. Figures appear not to sum exactly due to rounding.
Source: McKinsey Value Intelligence; McKinsey Global Institute analysis

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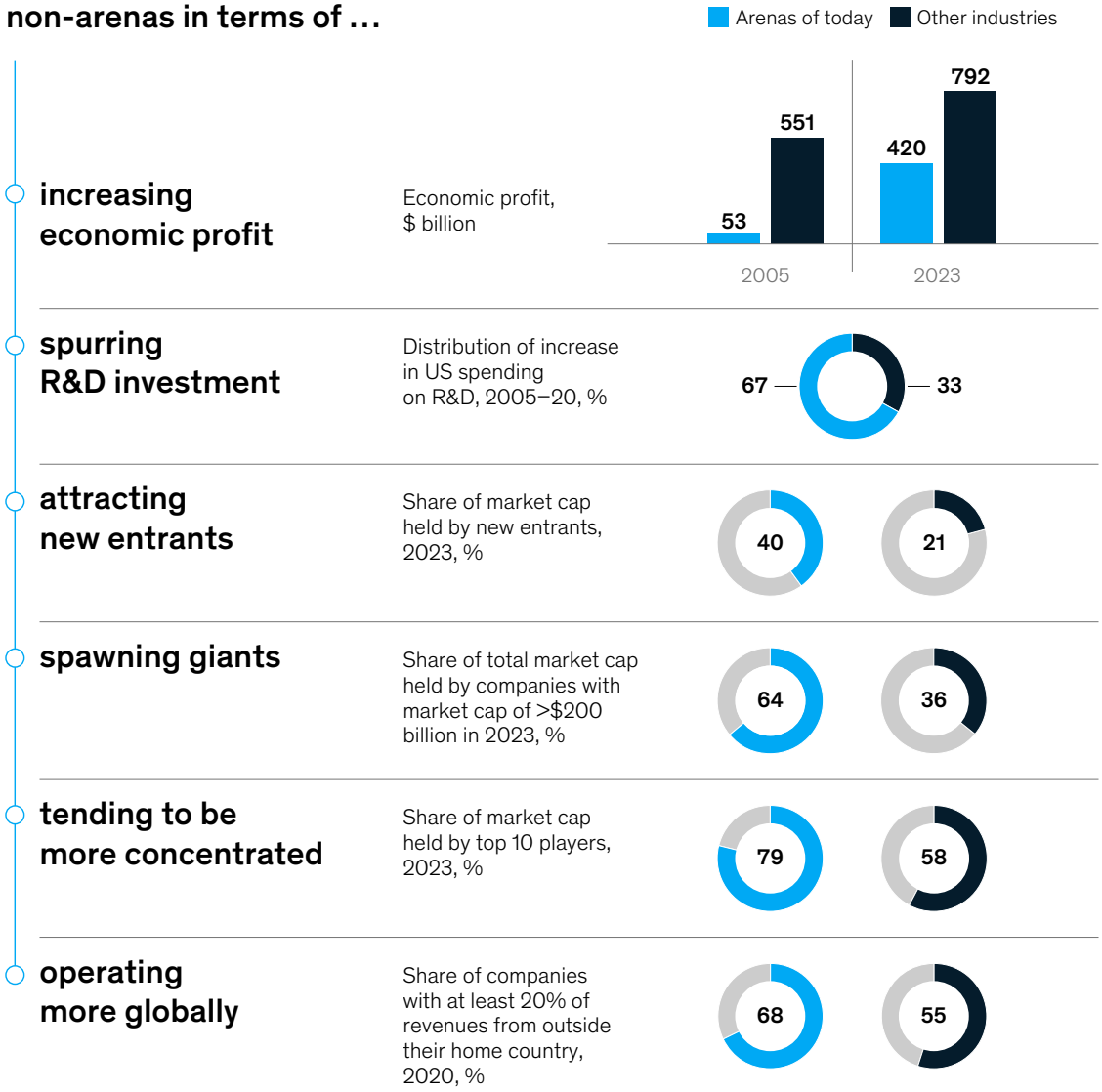
Six key metrics make today's arenas stand out

Beyond the defining features of growth and dynamism, much more is going on. Six metrics highlight how today's arenas stand out from other industries (Exhibit 4).

Arenas captured an increasing share of economic profit: Between 2005 and 2023, arenas increased their share of total global economic profit from 9 percent to 35 percent. This economic profit is revenue minus the costs of doing business and of capital. Leading arenas by economic profit in 2023 were consumer internet ($90 billion), consumer electronics ($76 billion), and semiconductors

Exhibit 4

Arenas of today stood apart from non-arenas in terms of …



Note: Exhibit and analysis updated to 2023 for four of the six metrics.
Source: McKinsey Global Institute analysis

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($62 billion), which benefited from rising smartphone penetration, cloud computing adoption, and the growing use of microchips. Remarkably, in 2024 Nvidia alone reached $66 billion in economic profit, more than the entire semiconductor industry in our sample in 2023. In contrast, many non-arena industries—including non-bank financials and telecom—experienced declining economic profits.

Arenas spur more R&D investment: Arenas received a disproportionate amount of R&D investment from 2005 to 2020. In the United States, about two-thirds of all R&D spending was allocated to arenas and arena-adjacent industries. As a share of revenues, arenas also spent more: In 2020, 10 percent of the revenues of arenas and arena-adjacent industries went to R&D, compared with 5 percent for other industries.[1]

Arenas attract new entrants: Arenas provide space for new players. In 2023, 40 percent of arena market capitalization belonged to companies that were either nonexistent or minor players in 2005, compared with just 21 percent in other industries. This influx of newcomers drives competition and innovation, thus increasing dynamism.

Arenas spawn giants: Arenas tend to include the world's largest companies. Sixty-four percent of arenas' total market cap was held by companies with market caps greater than $200 billion, compared with only 36 percent in other industries.

Arenas tend to be more concentrated: Market caps tend to become more concentrated in the arenas. Even though there were more new entrants in arenas than in non-arenas, in 2023 the top ten companies in each arena accounted for roughly 80 percent of its total market cap, versus only about 60 percent for non-arenas. Despite concentration, competitive pressure to innovate persists and regularly disrupts the ranks of winners.

Arenas operate more globally: On average, 50 percent of revenues in arenas were generated outside companies' home regions, compared with 42 percent for non-arena companies. Moreover, companies in arenas were much more likely to be multinationals.[2] Sixty-eight percent of companies from arenas derived more than 20 percent of their revenues from countries other than their own, compared with about half of companies in other industries.

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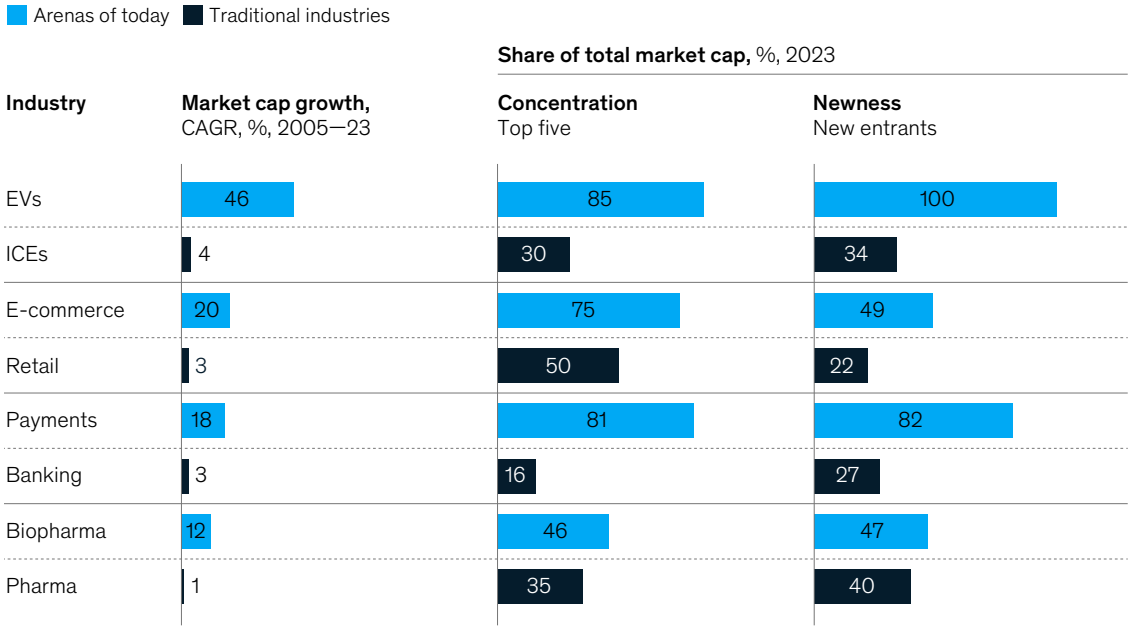
Arenas can emerge from and outgrow traditional industries

Some arenas are carved out of large established industries that don't exhibit outsize growth and dynamism. This gives us a natural set of comparable examples of traditional industries and arenas to illustrate these differences. In Exhibit 5, we display four of them: EVs and internal combustion engine (ICE) vehicles, e-commerce and brick-and-mortar retail, payments and banking, and biopharma and pharmaceuticals. The four relatively new industries are examples of arenas that have sufficiently novel technologies or business models to merit separating them from the industries they disrupted during this period.

Exhibit 5

Industry pairs demonstrate how arenas stand out.

Market cap growth and shares of total market cap, by industry pair



Source: McKinsey Value Intelligence; McKinsey Global Institute analysis

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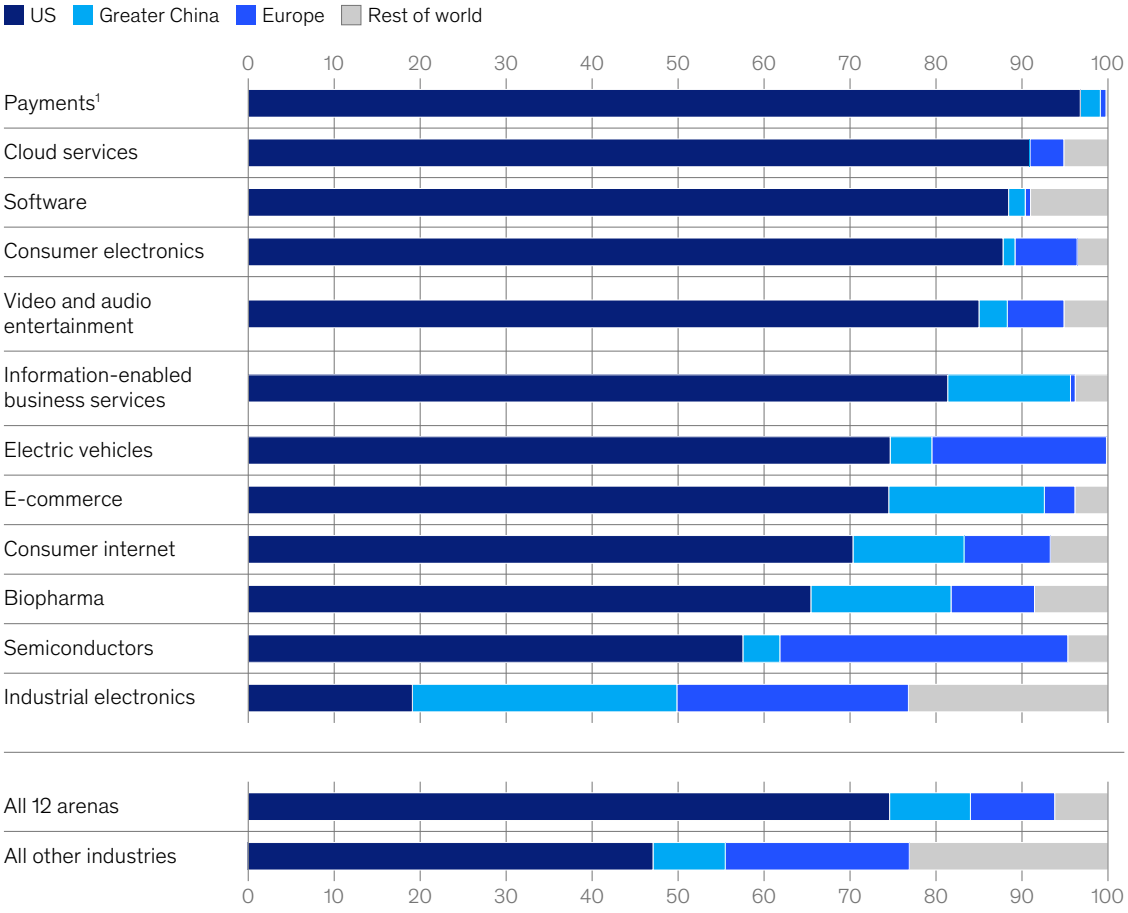
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US and Chinese companies are overrepresented in arenas, with US companies accounting for about three-fourths' share

In 2023, US-based companies accounted for most high-growth arenas, holding 75 percent of total market capitalization—up from about 65 percent since just 2020—and leading in all arenas except industrial electronics, where China led. Chinese companies also have a significant presence in the e-commerce, semiconductor, consumer internet, and EVs arenas—and, as a result, accounted for roughly 10 percent of market cap across all arenas. European companies accounted for substantial market cap in biopharma, industrial electronics, and information-enabled business services. However, Europe's overall market cap share was much higher in non-arena industries (21 percent) than in arenas (10 percent) (Exhibit 6).

Exhibit 6

Regional distribution of market cap by arena, 2023, %



Note: Companies' regions defined by their country of incorporation. Exhibit and analysis updated to 2023.
[1]Market cap information for China's leading payments player, Alipay, was unavailable. As a result, China's share of global payments is most likely undercounted.
Source: McKinsey Value Intelligence; McKinsey Global Institute analysis

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7

Three ingredients make up the arena-creation potion

To spot future potential arenas, we have identified three elements that, when present and combined, result in high growth and high dynamism, allowing for an arena to emerge. These three elements are referred to as the "arena-creation potion" and include the following:

— Business model or technology step changes

— Escalation mechanism for investments

— A large or growing addressable market

The first element is a significant step change in technology or business models that disrupts how products and services are developed or delivered. These shifts often follow an S-curve pattern, where performance or adoption starts slowly, accelerates rapidly after reaching an inflection point, and eventually levels off as maturity is reached. Real-world patterns may deviate from this idealized curve, but the framework remains useful for understanding transformation dynamics.

Business model step changes, often enabled by technology and innovation, can shift commercial models and disrupt market structures. For example, physical retail shifted to e-commerce, and video rentals to subscription-based streaming. Similarly, the semiconductor industry created a fabless business model over the past two decades, with some companies outsourcing the fabrication of physical microchips in order to focus on design and sales. Design-focused firms like Nvidia and Qualcomm emerged alongside manufacturing specialists like TSMC and equipment makers like ASML.[3]

The second element is the type of investments that players have an incentive to make, namely escalatory investments. Escalatory investments have two mutually compounding consequences, because increases in scale also enhance capabilities. These investments boost margins and expand or protect market share. Unlike basic ticket-to-play investments, such as opening new factories, escalatory investments give companies a competitive edge and deliver bigger and bigger returns as they grow.

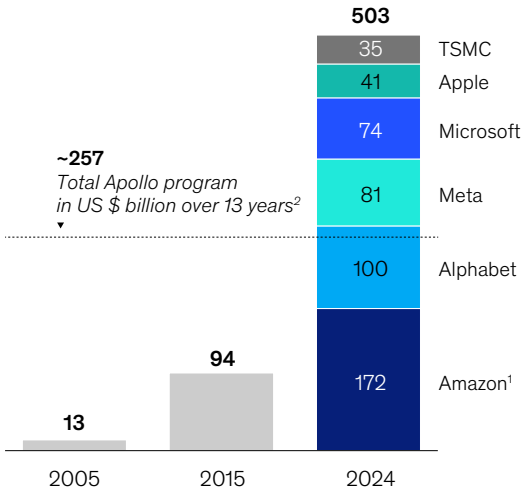
As a result, competitors also have a strong incentive to invest, beginning an arms race in which they iteratively invest to scale and scale to invest, causing a simultaneous escalation in capabilities. Investments may be directed to R&D, marketing, or strategic acquisitions, where returns grow over time and reinforce a firm's competitive position. For example, e-commerce platforms benefit from marketing investments that attract customers and drive network effects, while biopharma companies invest in AI-driven research to increase drug development success rates and ROI.

As escalatory investments continue and compound, spending levels are reaching new heights. Alphabet, Amazon, Apple, Meta, Microsoft, and TSMC—six major tech players competing across arenas that include cloud services, software, semiconductors, and e-commerce—collectively increased their capital expenditures and R&D investments from $13 billion in 2005 to $503 billion in 2024, growing at a 23 percent annual rate. To put that in perspective, the roughly $500 billion spent in one year could have funded the Apollo program, the original moonshot project spanning 13 years, twice over (Exhibit 7).

Exhibit 7

Investments by big tech players escalated nearly 40-fold.

Select companies' spending on capital and R&D, $ billion



Note: This analysis does not include M&A investment, only R&D and capital expenditures.
[1]Amazon's R&D spend refers to an operating expense reported as "technology and infrastructure" and may include expenses beyond what is typically reported as R&D.
[2]Adjusted for inflation from 1960–73 expenditures to 2020 US dollars; approximate figure.
Source: McKinsey Value Intelligence; McKinsey Global Institute analysis

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The third element—a large and often rapidly growing addressable market—may already exist or may be created by offering new products or service categories that displace existing markets.

In our analysis, biopharmaceuticals, industrial electronics, and software—which saw compound annual revenue growth rates of 6 to 13 percent from 2005 to 2023—generally addressed existing (and growing) markets. Over the same period, cloud services and e-commerce achieved 30 and 27 percent CAGR, respectively, with new offerings. Opening new markets often accelerates growth. For example, companies in the emerging EV arena grew market cap by about 50 percent CAGR from 2005 to 2023, while traditional automotive companies saw single-digit growth over the same period.

8

Eighteen potential arenas of the future are emerging

With our understanding of what makes an arena, we identified 18 potential arenas of the future. Some of them are existing arenas already tracing steep S-curves, while others are new and still in their early stages. We then grouped the 18 into five themes based on whether they are part of the foundation—including semiconductors, AI software and services, and cloud services—or are further downstream in the digital and physical realms, where AI catalysts now often enable technology resets (Exhibit 8).

The foundation: Includes semiconductors, AI software and services, and cloud services. Semiconductors power everything from smartphones to supercomputers. As computing demand grows, advancements in chip design, materials, and fabrication are key to unlocking next-gen capabilities in digitization and AI. Meanwhile, AI software and services are advancing and transforming every sector, with enterprise and consumer offerings. Cloud services are increasingly inseparable from AI as data storage and computing power continue to scale exponentially.

These three arenas are increasingly intertwined in an AI supercluster. The results are catalyzing and amplifying arenas that we have grouped under the themes of digitization, electrification, new bio-frontiers, and hard tech. While digitization is a supercharged version of the consumer-facing industries that were arenas in the 2005–23 period, the other three tend to be more enterprise- or industry-facing and have more physical dimensions.

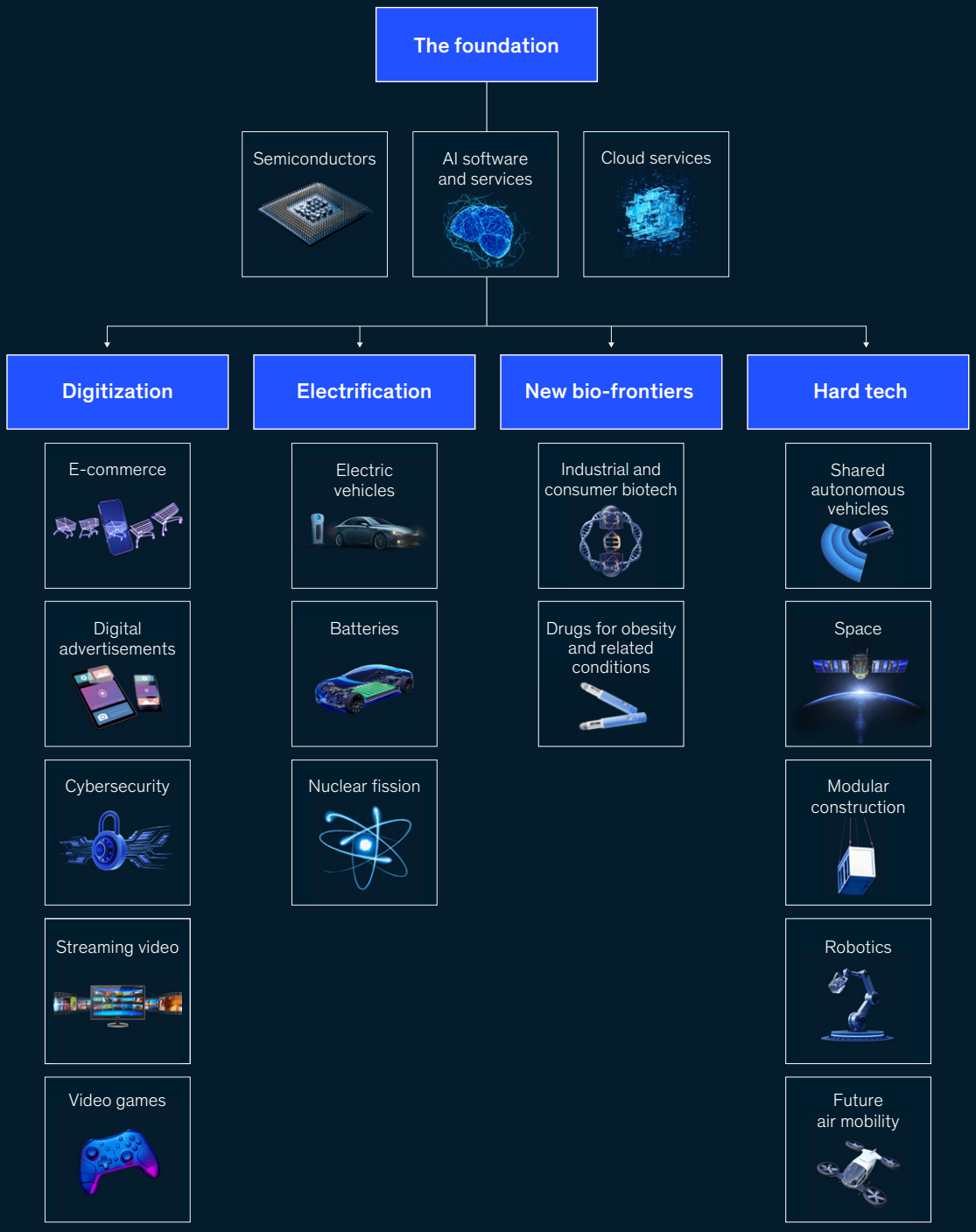
Digitization: Includes e-commerce, digital advertisements, cybersecurity, streaming video, and video games. As the foundations advance—increasingly enabled by AI tools that predict consumer behavior, for example—digitization technologies also advance and redefine how we shop, play, and consume content.

Electrification: Includes EVs, batteries, and nuclear fission. The world is being electrified, increasing energy efficiency, reducing energy emissions, and transforming how we power vehicles, devices, robots, and more.

New bio-frontiers: Includes industrial and consumer biotech, and drugs for obesity and related conditions. AI is applied to bio-frontiers, accelerating drug discovery and development, designing new synthetic biology and bioengineering, tailoring interventions for personalized and preventive health solutions, and improving the efficiency of biomanufacturing operations.

Hard tech: Includes shared autonomous vehicles, space, modular construction, robotics, and future air mobility. AI is automating and optimizing domains that manual labor and operations have traditionally dominated.

Exhibit 8



Source: McKinsey Global Institute analysis

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The 18 future arenas could generate $29 trillion to $48 trillion in revenues, or 18 to 34 percent of total GDP growth

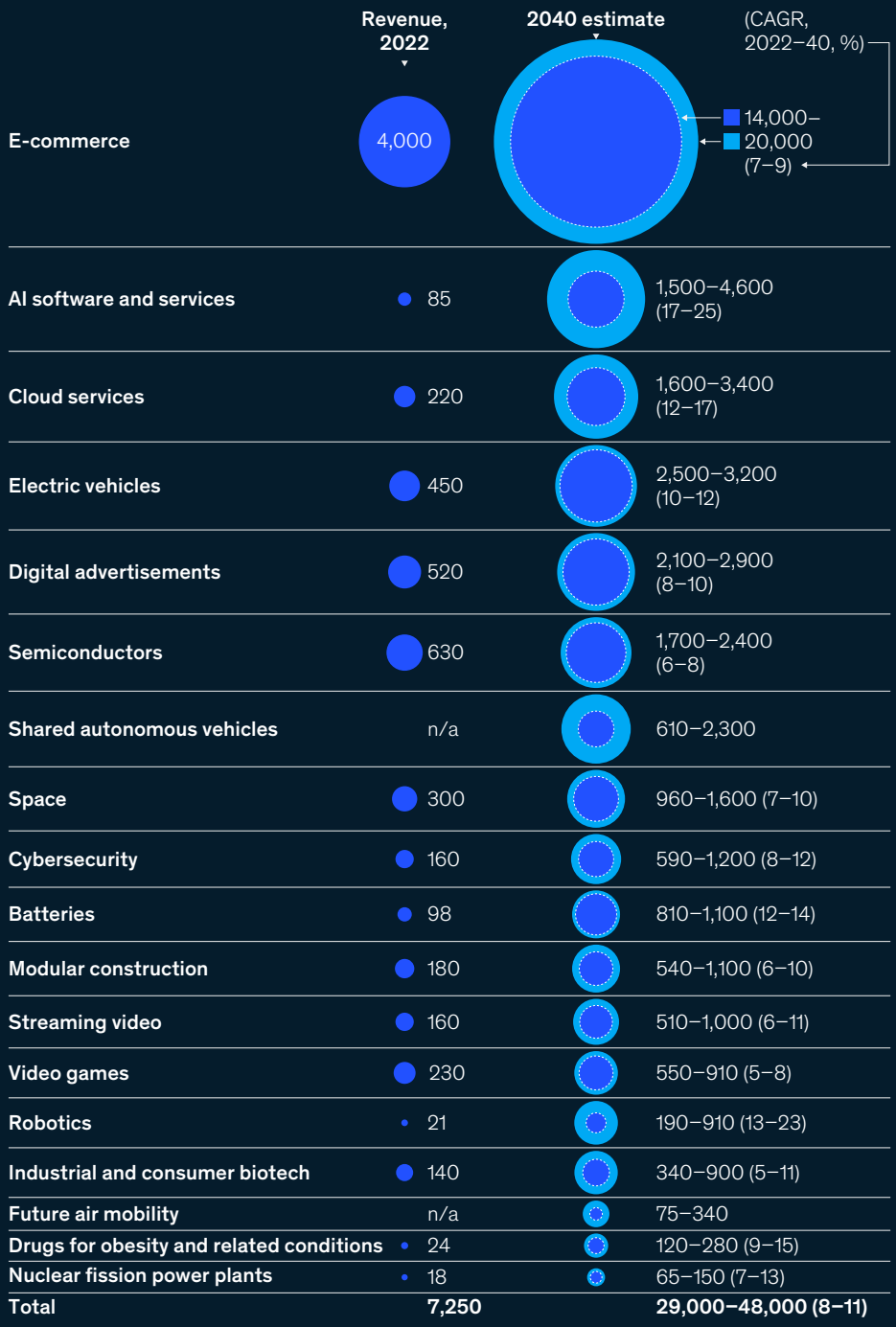
These 18 potential future arenas could yield a total of $29 trillion to $48 trillion in revenues and $2 trillion to $6 trillion in profits by 2040 (Exhibit 9). Converted into GDP terms, those revenues could represent 18 to 34 percent of total GDP growth.

While revenue and profit projections for future arenas generally align, variations in business models create notable differences in rankings by profit potential. For instance, based on these industries' profit margins today, the semiconductor arena could generate profit margins of 20 to 25 percent, making it the third-largest arena by potential 2040 profit, though it is only the sixth-largest arena by 2040 revenue. By contrast, the EV arena, which ranks fourth by 2040 revenue (also ranked by the higher range of estimates), would rank seventh by potential 2040 profit if the industry's profit margins stay between 4 and 10 percent.

While revenue and profit projections for future arenas generally align, variations in business models create notable differences in rankings by profit potential.

Exhibit 9

18 potential arenas of tomorrow, by 2040 revenue estimate, $ billion



Source: Company annual reports; McKinsey Value Intelligence; McKinsey Global Institute analysis

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10

We are all in the arena now

Given arenas' outsize growth and dynamism, companies and investors should understand their own exposure and positioning. Arenas cannot be ignored. A practical first step is to activate "arenas radar": Scan for emerging opportunities and risks, and measure the value at stake versus a passive stance (Exhibit 10). Start by assessing whether your company can compete in or is likely to be disrupted by any of the arenas. Then look for opportunities to use arena products and services to reconfigure production capabilities or tap into downstream revenue streams that arenas may be creating.

The following questions illustrate what a scan entails.

— *Are you positioned to compete (or be disrupted)?* Are you an active participant in any arenas already? Are there arenas where you have the technology, human capital, operations, or other capabilities to compete soon? For example, retailers have transformed into e-commerce players over the past two decades and are becoming players in digital advertisements.

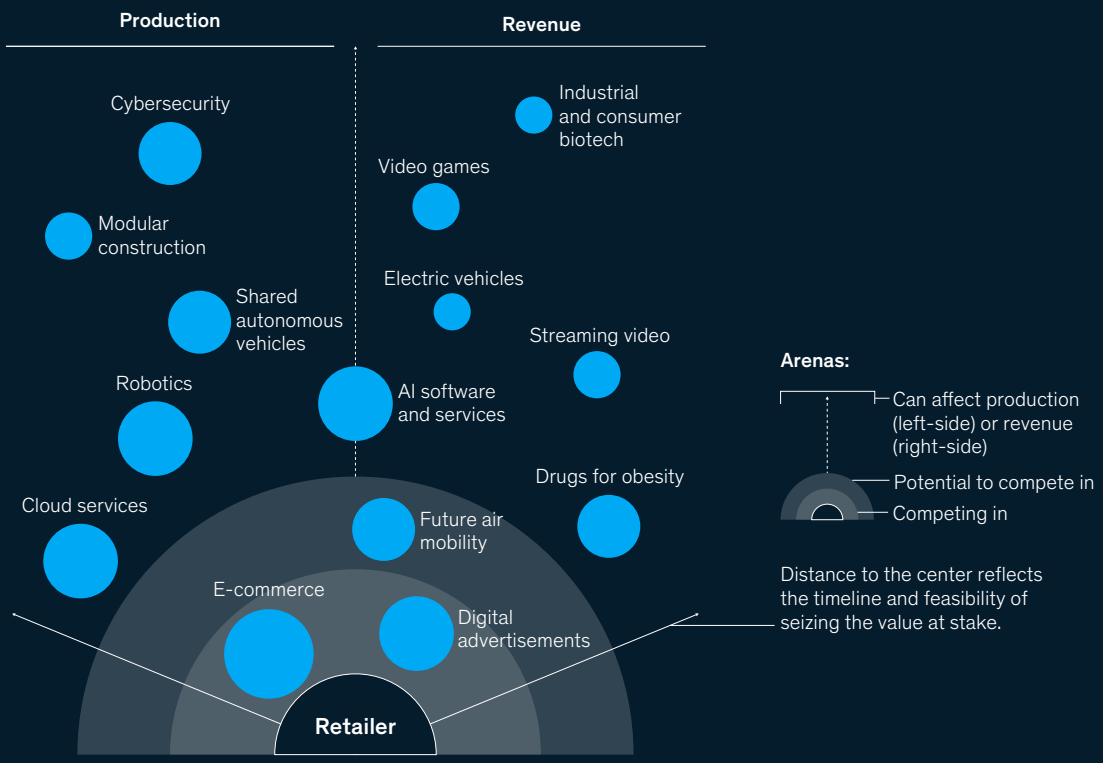
— *Are you a potential customer?* What new capabilities can you adopt and accelerate? For example, a consumer packaged goods retailer could use the products developed in arenas, like deploying robotics to become more efficient in its store operations or drones for delivery.

— *Are you a supplier?* What are the new opportunities—and risks—to your downstream demand? For example, an industrial machinery supplier producing robotics parts may see that segment of its business grow rapidly, while also needing to make sure that it stays on the leading edge of robotics advancements to ensure that its parts remain critical.

— *Are you an investor?* How can you catch the waves at the right time? For example, a private equity company investing in more nascent arena industries—like future air mobility, shared autonomous vehicles, or robotics—would have to consider when these markets would be likely to see high growth and adoption as well as how profits along the value chain would be distributed.

Exhibit 10

Retailer example



Source: McKinsey Global Institute analysis

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Our analyses of present and future arenas reveal three key swing factors that could be fundamental to the evolution of the arenas of tomorrow in technology, investment patterns, and sources of demand. These swing factors are (1) geopolitical developments affecting the regulation of innovation and technological regionalization; (2) AI advances and adoption in a range of industries; and (3) the green transition, which aims to alter the course of climate change and could drive demand in various parts of the market.

This article provides a condensed view of where to expect the most growth and dynamism—and how to update that view as the future takes shape. In the full report, a compendium sketches the quantitative possibilities for the range of growth and dynamism prospects of each of the 18 potential arenas of tomorrow. Two additional articles, on future growth and dynamism in New York and India, explore how these globally transformative industries matter to cities and regions at different levels of economic development.

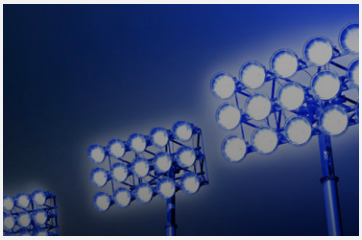
Looking ahead, the 18 arenas of tomorrow could be even more materially transformative than the 12 arenas of the past couple of decades, shaping how we consume and process data, approach health and wellness, and interact and communicate with one another. They could introduce new options for our lives and raise further questions about our social progress, from the moral and ethical questions surrounding data and privacy to imperatives for businesses to be inclusive and sustainable. Recognizing how and when arenas originate, understanding how they evolve, and anticipating the ways in which they could change society can offer valuable foresight. Arenas of today and tomorrow are closer and more consequential than many believe. With a better understanding of arenas, leaders can anticipate—and act on—the accelerating pace of change.

Chris Bradley is a McKinsey senior partner in the Sydney office and a director of the McKinsey Global Institute (MGI), **Kevin Russell** is an MGI senior fellow based in the Charlotte office, **Kweilin Ellingrud** is a McKinsey senior partner based in the Minneapolis office and a director of MGI, and **Suhayl Chettih** is an MGI fellow based in the New York office.

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For additional research on these topics, please explore the following materials:



Report - *MGI Research*

The next big arenas of competition

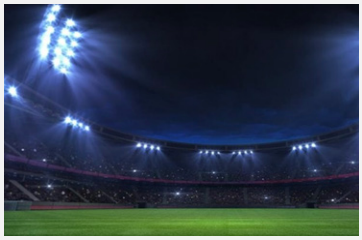
Arenas are industries that transform the business landscape. Eighteen future arenas could reshape the global economy and generate $29 trillion to $48 trillion in revenues by 2040. Includes full compendium on arena deep dives.



Article

New York 2040: Time to reinvent. Again.

In this ever-evolving region, 18 arenas of competition will present challenges and opportunities that determine the future of New York.



Article

India's future arenas: Engines of growth and dynamism

India, which played a limited role in the global arenas before, could now build on inherent endowments and large-scale sectors to excel in 18 high-growth, dynamic arenas—global and national—and generate up to $2 trillion in revenues by 2030.

Endnotes

1 National Center for Science and Engineering Statistics and Census Bureau, Business Enterprise Research and Development Survey, 2020; McKinsey Global Institute analysis.

2 Home regions are defined by the location of the company's headquarters.

3 Kasper Karlsson, "Understanding the semiconductor value chain," Quartr, March 21, 2024.

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